

HOMESTEAD AFB, FL, USA

WMO: 722026

Lat: 25.483N

Lon: 80.383W

Elev: 5

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 12826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.1	50.3	33.8	28.4	53.8	39.0	34.9	54.9	23.6	66.9	20.6	67.8	6.8	340	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	91.3	79.3	90.4	79.1	89.7	78.9	81.6	87.4	80.9	86.9	80.2	86.5	9.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.4	157.9	84.5	79.2	151.7	83.8	78.6	148.7	83.6	45.1	88.0	44.2	87.0	43.5	87.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	18.2	16.5	DB	40.0	93.8	4.1	1.6	37.1	95.0	34.7	95.9	32.5	96.9	29.5	98.1	(4)
(5)			WB	37.1	82.6	5.0	1.1	33.6	83.3	30.7	84.0	27.9	84.6	24.3	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.3	67.0	69.7	72.0	75.2	78.7	81.7	82.9	83.1	82.0	78.9	73.6	70.4
	DBStd	6.98	6.94	6.14	5.32	3.90	2.58	2.08	1.91	1.76	1.86	3.59	4.71	6.47
	HDD50	2	1	0	0	0	0	0	0	0	0	0	0	1
	HDD65	144	64	29	14	1	0	0	0	0	0	0	5	31
	CDD50	9602	528	551	682	757	890	952	1019	1027	961	896	707	632
	CDD65	4269	126	159	231	308	425	502	554	562	511	432	261	198
	CDH74	38298	508	794	1340	2182	3713	5269	6292	6439	5368	3862	1571	960
	CDH80	11941	23	80	227	459	1021	1916	2504	2552	1850	1011	219	79
Wind	WSAvg	7.4	7.8	7.7	8.6	8.6	8.2	6.3	5.7	5.9	6.4	8.4	8.1	7.5
Precipitation	PrecAvg	58.1	1.9	2.0	2.1	2.8	5.3	9.4	6.9	8.3	8.5	6.1	2.5	2.1
	PrecMax	72.4	6.0	5.8	5.8	5.5	12.5	18.6	10.8	12.3	13.1	14.6	8.0	12.4
	PrecMin	48.1	0.4	0.0	0.3	0.3	0.6	3.1	3.7	3.3	3.7	1.5	0.4	0.3
	PrecStd	6.9	1.3	1.4	1.3	1.4	2.8	4.2	2.1	2.4	2.5	3.6	2.2	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.1	84.4	87.1	88.7	90.4	92.7	93.0	92.5	91.4	89.6	86.4	83.9
		MCWB	73.2	73.8	74.2	75.3	77.2	79.6	79.7	79.6	78.7	78.2	76.6	75.8
	2%	DB	80.7	82.1	84.3	86.4	88.2	90.7	91.0	91.1	90.1	87.8	84.2	82.0
		MCWB	72.6	73.2	73.9	74.6	76.8	79.1	79.4	79.6	78.8	78.0	75.9	74.5
	5%	DB	79.0	80.8	82.2	84.3	86.4	89.5	90.1	90.2	88.6	86.3	82.3	80.9
		MCWB	71.6	72.5	72.8	73.9	76.2	78.7	79.2	79.4	78.7	77.6	74.3	73.7
	10%	DB	77.1	79.0	80.9	82.4	85.1	88.0	89.0	89.2	87.8	85.2	81.0	79.2
		MCWB	70.1	71.4	72.0	72.9	75.7	78.3	78.9	79.2	78.6	77.1	73.1	72.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.6	77.0	78.6	78.2	80.5	82.8	82.4	82.7	81.6	81.2	78.8	78.1
		MCDB	79.4	80.5	82.3	84.6	85.7	88.2	88.9	89.1	86.5	85.0	82.1	80.5
	2%	WB	74.4	75.3	75.8	76.8	78.6	81.2	81.2	81.4	80.8	79.6	77.4	76.4
		MCDB	78.0	79.4	81.6	83.1	84.6	86.9	87.9	87.7	86.1	84.8	82.1	79.7
	5%	WB	73.1	74.1	74.4	75.6	77.7	80.2	80.4	80.5	80.1	78.8	76.1	75.1
		MCDB	77.4	78.6	80.0	81.7	84.0	86.3	87.2	87.1	85.8	84.0	80.8	78.9
	10%	WB	71.6	72.8	73.1	74.6	76.7	79.2	79.7	79.9	79.4	78.1	75.0	74.0
		MCDB	75.8	77.7	78.6	80.6	83.0	85.6	86.7	86.8	85.4	83.2	79.5	77.9
Mean Daily Temperature Range	5% DB	MDBR	16.1	15.6	15.1	14.0	12.9	12.6	12.6	12.4	11.6	11.5	13.7	14.3
		MCDBR	14.3	15.0	16.0	16.0	15.1	14.5	13.8	13.6	12.9	11.9	12.7	12.6
		MCWBR	8.5	8.0	7.7	7.4	7.0	6.9	6.2	5.7	5.5	5.2	6.2	7.0
	5% WB	MCDBR	13.2	13.7	14.8	13.8	13.6	13.2	12.8	12.7	11.9	11.0	11.5	11.4
		MCWBR	8.6	7.7	8.1	7.2	7.0	6.7	6.5	6.0	5.6	5.5	6.2	7.1
Clear-Sky Solar Irradiance	taub	0.363	0.378	0.387	0.399	0.427	0.457	0.464	0.452	0.432	0.417	0.382	0.375	
	taud	2.536	2.503	2.465	2.415	2.383	2.360	2.331	2.394	2.442	2.452	2.500	2.519	
	Ebn at Noon	283	287	289	286	276	266	264	267	271	270	274	274	
	Edh at Noon	29	32	35	38	39	40	41	38	36	34	30	29	
All-Sky Solar Radiation	RadAvg	1191	1437	1762	1984	1984	1800	1858	1793	1579	1453	1254	1110	
	RadStd	76	79	75	91	155	154	73	93	80	101	56	82	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	+0.68	+1.58	N/A	N/A	N/A	N/A	N/A	-31	+232	+193

Nomenclature: See separate page